

Unsupervised Learning on Graphs (GAE) and Heterogeneous Graph Auto-Encoders for Credit Card Fraud Detection: An In-Depth Analysis and Review

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Unsupervised representation learning on graphs, particularly through Graph Auto-Encoders (GAEs), has emerged as a powerful paradigm for learning from data without exhaustive human annotation. It is especially attractive in domains dominated by extreme class imbalance, such as financial fraud detection, where anomalous instances are exceedingly rare. This paper presents a critical review of unsupervised graph representation learning, taking the Heterogeneous Graph Auto-Encoder (HGAE) of Majumder et al. for credit card fraud detection as a primary case study. We examine how the HGAE models cardholders, merchants, and transactions as a heterogeneous graph, learns latent representations of legitimate behaviour, and flags fraud as anomalies through reconstruction error, thereby circumventing the need for fraud labels during training. Through snowball research, we contextualise the model within the broader evolution of self-supervised graph learning—spanning attention-based heterogeneous networks, expressiveness results for simplified GAEs, and masked and contrastive objectives—and provide a detailed critical analysis of its architectural, methodological, and evaluative limitations. Finally, we outline future research directions, including temporal heterogeneous modelling, contrastive regularisation, and scalable linear propagation, aimed at advancing robust unsupervised fraud detection.

Graph Auto-Encoders | Fraud Detection | Heterogeneous Graphs | Unsupervised Learning | Anomaly Detection

The digital revolution has catalyzed an unprecedented surge in digital transactions, bringing convenience alongside a sophisticated ecosystem of financial fraud. Traditionally, supervised machine learning (ML) models have served as the first line of defense. However, supervised paradigms face a fundamental bottleneck: the extreme class imbalance phenomenon. In real-world datasets, legitimate transactions overwhelmingly dominate (e.g., 1,842,743 normal transactions versus a mere 9,651 fraudulent instances—a $\sim 191:1$ ratio) Majumder et al. (2025). Supervised methods often require heuristic resampling (like SMOTE), which induces severe overfitting or structural information loss Majumder et al. (2025).

To transcend these limitations, the academic focus has shifted toward *Unsupervised Learning on Graphs*. Unsupervised methods, specifically Graph Auto-Encoders (GAEs), generalize standard autoencoder theory to non-Euclidean domains Kipf and Welling (2016). They offer an elegant solution to the label-scarcity problem: by training a model to reconstruct the attributes of legitimate transaction nodes—while exploiting the graph topology during encoding—any deviation (high reconstruction error) is inherently flagged as an anomaly Majumder et al. (2025).

Furthermore, modern financial networks are inherently heterogeneous, comprising distinct entity types (cardholders, merchants, transactions). This paper investigates the Heterogeneous Graph Auto-Encoder (HGAE) as a prime manifestation of unsupervised graph learning. We critically dissect its architecture, contextualize it within the broader evolution of self-supervised graph models via snowball research, and explore how recent findings on GAE expressiveness and contrastive objectives can shape future fraud detection systems.

1. Background: Unsupervised Learning on Graphs

To systematically contextualize the HGAE framework, we trace its theoretical lineage through the evolution of graph representation learning, moving from supervised message passing toward unsupervised and self-supervised auto-encoding.

A. From Message Passing to Attention. Modern Graph Neural Networks are unified under the *message-passing* paradigm: each node iteratively refines its representation by aggregating transformed messages from its neighbours. The Graph Convolutional Network (GCN) realises this through a normalised adjacency operator Kipf and Welling (2017), while

Significance Statement

Credit-card fraud detection is hampered by extreme class imbalance: fraudulent transactions are vanishingly rare, so supervised classifiers require aggressive resampling that distorts the data. This review examines an unsupervised alternative—the Heterogeneous Graph Auto-Encoder—that learns only the behaviour of legitimate transactions and flags fraud as reconstruction anomalies, sidestepping the need for fraud labels. By tracing the model's lineage through attention-based heterogeneous networks and modern self-supervised graph learning, and by critically dissecting its architectural and evaluative assumptions, we clarify when graph auto-encoders genuinely help and when their advantages are overstated. The analysis gives practitioners a grounded view of unsupervised graph learning for anomaly detection and charts concrete directions—temporal modelling, contrastive regularisation, and scalable linear propagation—for more robust fraud detection.

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GraphSAGE introduces inductive neighbour sampling to scale aggregation to massive graphs [Hamilton et al. \(2017\)](#). The Graph Attention Network (GAT) replaces fixed aggregation coefficients with a learned attention function, enabling a node to weight neighbours by relevance [Velićković et al. \(2018\)](#). A defining theoretical property shared across these models is *permutation equivariance*: the output transforms consistently under any re-ordering of nodes, the inductive bias that distinguishes graph learning from grid- or sequence-based architectures. Two well-known constraints, however, shape practical design: the discriminative power of message-passing GNNs is upper-bounded by the 1-Weisfeiler–Lehman isomorphism test [Xu et al. \(2019\)](#), and stacking many layers triggers *oversmoothing*, where node embeddings converge toward indistinguishability. These constraints directly motivate both the shallow-yet-expressive designs we discuss below and several of our later criticisms.

B. Heterogeneous Graphs and the Graph Transformer. Real financial networks are inherently *heterogeneous*: they comprise multiple node types (cardholders, merchants, transactions) connected by multiple relation types. Applying a homogeneous GNN to such data collapses these semantic distinctions. The Heterogeneous Graph Transformer (HGT) resolves this by maintaining *type-specific* projections and *relation-specific* mutual attention: it computes query, key, and value vectors in the spirit of the Transformer [Vaswani et al. \(2017\)](#), but parameterises them per node and edge type [Hu et al. \(2020\)](#). The HGAE case study [Majumder et al. \(2025\)](#) constructs its encoder directly upon this HGT formulation, inheriting its multi-head, relation-aware attention; recognising this lineage is essential for evaluating which aspects of the HGAE are genuinely novel versus inherited.

C. The Paradigm of Graph Auto-Encoders. GNN-based auto-encoders constitute a broad class of self-supervised models that compress graph-structured data into compact latent representations [Kipf and Welling \(2016\)](#); [Li et al. \(2024\)](#). The classical Graph Auto-Encoder (GAE) couples a GNN encoder, which maps node features and adjacency into embeddings Z , with a decoder that reconstructs the graph. In the canonical formulation the decoder is a parameter-free inner product, $\hat{A} = \sigma(ZZ^\top)$, so that all learning capacity resides in the encoder and the reconstruction target is the *topology* itself. The Variational Graph Auto-Encoder (VGAE) extends this by imposing a Gaussian prior on the latent space and optimising the Evidence Lower Bound (ELBO), whose Kullback–Leibler term is precisely what constrains the posterior and makes the model *variational* [Kipf and Welling \(2016\)](#). Crucially for our case study, such models operate without ground-truth anomaly labels: by learning to reconstruct only the “normal” state of a system, they recast detection as a one-class problem in which anomalies surface as high reconstruction error. This reconstruction-as-anomaly-score principle is the conceptual core that the HGAE adapts to fraud detection.

D. Reconsidering GAE Performance and Expressiveness. Historically, standard GAEs were criticized for limited expressiveness, particularly their inability to distinguish certain non-isomorphic structural contexts [Ma et al. \(2025\)](#). Recent evidence, however, demands a re-evaluation. A pivotal 2025 study by Ma et al. [Ma et al. \(2025\)](#) demonstrated that a

heavily optimised, deliberately *simple* GAE can match or surpass far more complex GNNs in link prediction. By applying orthogonal initialization to learnable embeddings and *removing* the non-linear activations between intermediate message-passing layers, the encoder is shown to implicitly perform pairwise neighbourhood (common-neighbour) modelling, attaining a state-of-the-art Hits@100 of 78.41% on the large-scale ogbl-ppa benchmark [Ma et al. \(2025\)](#). This result reinforces the premise of our case study: a well-designed unsupervised auto-encoder is a potent instrument for structural anomaly detection, and architectural restraint can be a feature rather than a deficiency.

E. Heterogeneous Graphs in Industry: The GEM Model. The transition from homogeneous to heterogeneous graphs is a decisive leap for industrial fraud detection. A seminal unsupervised/semi-supervised work in this domain is the GEM model deployed at Alipay [Liu et al. \(2018\)](#). GEM operates on the premise of attackers’ structural weaknesses—*Device Aggregation* and *Activity Aggregation*—and uses an attention mechanism over a heterogeneous graph of accounts and devices (MAC, IP, phone numbers) to adaptively learn the importance of different node types [Liu et al. \(2018\)](#). The HGAE framework for credit card fraud [Majumder et al. \(2025\)](#) conceptually mirrors this philosophy: it evaluates distinct entity semantics and reconstructs the behaviour of legitimate topologies, but differs in framing detection purely as attribute reconstruction error rather than supervised classification over labelled accounts.

2. Methodology and Architectural Findings

The HGAE architecture mitigates the complexities of multi-relational financial data through a series of mathematically rigorous mechanisms. Formally, the financial network is defined as a Heterogeneous Graph $G = (V, E, T, R, X)$, where V and E denote nodes and edges, T and R signify the sets of node types (Customer, Merchant, Transaction) and edge types, and X represents the attribute matrix [Majumder et al. \(2025\)](#). We first describe how raw records become this graph, then detail the encoder, the probabilistic latent space, and the decoder-based detection rule. Figure 1 summarises the end-to-end pipeline, from a tabular transaction log to a binary fraud decision.

A. Data Representation and Preprocessing. The model consumes a tabular transaction log, which must first be lifted into a heterogeneous graph. Each transaction record instantiates (or connects to) three typed nodes: a *Customer* node (keyed by `cc_num`), a *Merchant* node (keyed by `merchant_id`), and a *Transaction* node. These are linked by typed edges, e.g. Customer $\xrightarrow{r_1}$ Transaction and Transaction $\xrightarrow{r_2}$ Merchant, as illustrated in Figure 2.

Two principles govern this construction. First, the identifier fields (`cc_num`, `merchant_id`, `trans_num`) serve only to define node identity and connectivity; they are *not* used as features, since raw identifiers carry no metric meaning. Second, the `is_fraud` label is excluded from the input entirely—it is reserved solely for threshold calibration and final evaluation, consistent with the unsupervised training regime.

Node attributes are then numerically encoded: continuous fields (transaction amount, timestamp-derived features, geographic coordinates) are standardised to zero mean and

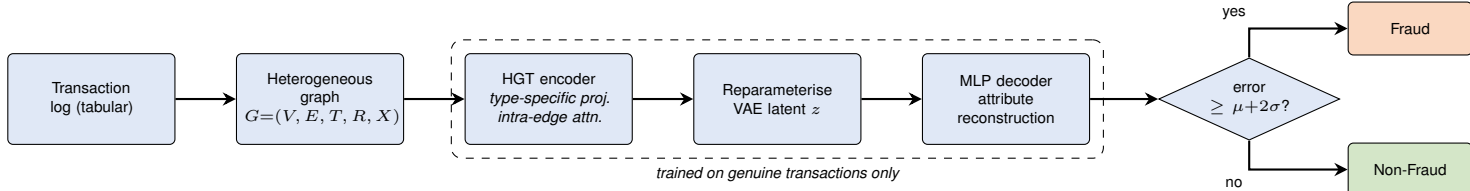


Fig. 1. The Heterogeneous Graph Auto-Encoder pipeline. A tabular log is lifted into a typed graph; an HGT-style encoder with type-specific projections and intra-edge attention produces a variational latent code; an MLP decoder reconstructs node *attributes*. Because the auto-encoder is trained on genuine transactions only, a reconstruction error exceeding the $\mu + 2\sigma$ threshold flags an anomaly (fraud).

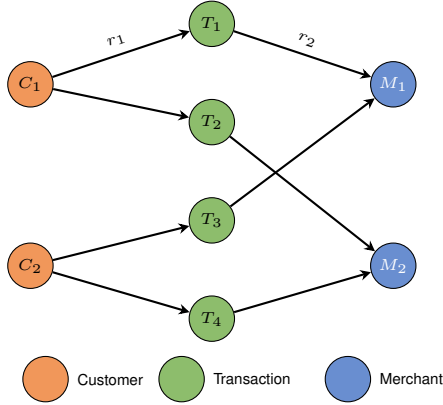


Fig. 2. A heterogeneous transaction graph with three node types. Customers (r_1) initiate transactions that involve merchants (r_2); a customer may transact with several merchants. Relation-specific edges let the encoder treat a Customer-to-Transaction link differently from a Transaction-to-Merchant link.

unit variance, categorical fields (e.g. gender, merchant category) are one-hot encoded, and missing values are imputed by the per-type mean or mode. The concatenation of these encoded attributes forms the initial node representation h^0 , the layer-zero input to the encoder. Because the three node types possess heterogeneous attribute sets—and therefore vectors of differing dimensionality and semantics—they are reconciled into a shared latent space by the type-specific linear projections introduced below.

B. Type-Specific Projections and Intra-Edge Attention. A profound architectural distinction of the HGAE compared to standard GAT models is its approach to attention. In conventional GAT, the Softmax function normalizes attention scores across all neighbors, forcing different entities to compete. In a heterogeneous financial graph, forcing a "Customer" node to compete with a "Merchant" node is semantically flawed.

To solve this, HGAE employs an *intra-edge* multi-head attention mechanism. First, nodes are mapped to a unified latent space via type-specific linear projections $Linear_t^{S_k}$, $Linear_t^{D_k}$, and $Linear_t^{M_k}$. For a specific edge e_r connecting a source v_t and destination d_t , the model utilizes H attention heads. The attention coefficient is computed as:

$$Att^k(v_t, e_r, d_t) = Linear_t^{S_k}(h_{v_t}) \cdot W_r^{Att} \cdot (Linear_t^{D_k}(h_{d_t}))^T \quad [1]$$

Crucially, the Softmax normalization is applied *across the H heads of a single edge*, not across the neighborhood:

$$f_{Attend} = \text{Softmax}(\|_{k \in [1, H]} Att^k(v_t, e_r, d_t)) \quad [2]$$

This allows the model to uniformly aggregate the total volume of messages from all neighbors using an element-wise addition operator ($\oplus$), but dynamically filters which specific connection subspace (head) is most informative for that specific connection Majumder et al. (2025). Each relation type r owns its own learnable matrices W_r^{Att} and W_r^{Msg} , so that the semantics of a Customer-to-Transaction link are encoded differently from those of a Transaction-to-Merchant link.

C. Reparameterization and Probabilistic Latent Space. To prevent the model from learning rigid, deterministic vectors that are highly susceptible to overfitting—especially given the sparse nature of fraud—the HGAE incorporates a reparameterization trick ($f_{reparam}$) at the final encoder layer.

$$h_{d_t}^l = \text{mean}(h_{d_t}^l) + \epsilon \cdot \exp\left(\frac{1}{2} \cdot \log(h_{d_t}^l)\right) \quad [3]$$

where $\epsilon \sim \mathcal{N}(0, 1)$. By treating each embedding as a Gaussian random variable parameterised by a learned mean and log-variance, the encoder constructs a continuous, smooth latent space, and the trick keeps the sampling operation differentiable so that gradients propagate through it during training.

D. Decoder Reconstruction and Anomaly Thresholding. The decoder is a Multi-Layer Perceptron (MLP) equipped with type-specific parameters. Importantly, it reconstructs each node's *attributes* rather than the graph adjacency: continuous features (e.g. transaction amount) are reconstructed under a Mean Squared Error (MSE) objective, and categorical features under Cross-Entropy. The training paradigm is entirely *unsupervised* (self-supervised): the objective depends solely on attribute reconstruction, devoid of any 'Fraud/Normal' labels during optimization:

$$L = \sum_{\forall N} \sum_{\forall t} \text{LOSS}(d_t, d'_t) \quad [4]$$

The end-to-end procedure thus separates into three phases. In *training*, the auto-encoder sees only legitimate transactions and learns to reconstruct the attribute patterns of normal financial behaviour, optimising L by back-propagation through both decoder and encoder. In *calibration*, reconstruction errors are computed over a held-out set of legitimate transactions and the decision boundary is statistically derived:

$$\text{Threshold} = \mu + 2\sigma \quad [5]$$

where μ and σ are the mean and standard deviation of those errors, corresponding to a $\sim 95\%$ interval for normal classification. Figure 3 depicts the resulting decision rule: genuine transactions cluster at low reconstruction error, whereas fraudulent

ones—never modelled during training—produce a heavier right tail that crosses the threshold. In *inference*, a new transaction is passed forward through the frozen network;

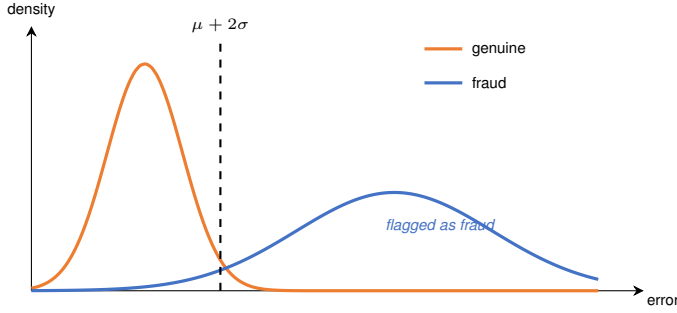


Fig. 3. Anomaly detection as reconstruction error. Training on genuine transactions concentrates their error to the left of the threshold $\mu + 2\sigma$; fraudulent transactions reconstruct poorly and fall to the right, where they are flagged. (Illustrative densities, following the loss histogram reported in the source paper.)

a reconstruction error below the threshold yields “Non-Fraud”, and one above it yields “Fraud”. The intuition is that, having never modelled fraudulent behaviour, the decoder reconstructs out-of-distribution transactions poorly, producing a surging loss. The overall time complexity is approximately $O(nE)$, with n the number of nodes.

Empirically, the model attains an AUC-PR of 0.89 and an F1-score of 0.81 (Table 1). On AUC-PR it edges out GraphSAGE (0.86) and FI-GRL (0.84). It is important to note, however, that its ROC-AUC (0.85) is in fact *lower* than these baselines (0.91–0.93), indicating a precision–recall trade-off under class imbalance rather than uniform superiority Majumder et al. (2025). We return to this discrepancy in our critical analysis (Section 3).

Table 1. Reported performance of graph-learning models on the credit-card fraud benchmark (source paper, Table 4). Best value per column in **bold**. AUC-PR is the most informative metric under the extreme class imbalance.

Model	AUC-PR	F1-score	ROC-AUC
Proposed HGAE	0.89	0.81	0.85
GraphSAGE + XGBoost	0.86	0.80	0.93
FI-GRL	0.84	0.70	0.92
Baseline	0.84	0.74	0.91

3. Criticism and Limitations

While the HGAE establishes a formidable benchmark, a critical, deep-dive evaluation exposes several structural, methodological, and evaluative constraints that must be addressed before deployment in mission-critical, hyper-scale financial infrastructures. We organise them into (i) formulation and reproducibility issues specific to this work, and (ii) limitations inherited from the broader GNN family.

A. Formulation and Reproducibility Concerns.

- **Implausible Reported Depth:** The reported configuration specifies 124 encoder layers. Such depth is fundamentally incompatible with the oversmoothing dynamics

of message passing: repeated neighbourhood averaging drives node representations toward a common fixed point, and embeddings would become practically indistinguishable far before the 124th layer. This figure is therefore most likely a typographical error or an unreported architectural deviation (e.g. heavy residual or sampling sub-structure); either way, it prevents faithful reproduction and is internally inconsistent with the model’s own reliance on discriminative embeddings.

- **Attribute Reconstruction Weakens the Graph Rationale:** Unlike the classical GAE, whose inner-product decoder reconstructs the adjacency structure, the HGAE decoder reconstructs node *attributes*. Consequently the relational signal—the very motivation for adopting a graph representation—enters the anomaly score only *indirectly*, through whatever the encoder has folded into the embedding, and is never explicitly verified by the reconstruction objective. A fraud pattern that is structurally anomalous but attribute-typical could therefore evade detection, undercutting the paper’s central argument that topology is what traditional methods miss.
- **Incomplete Variational Formulation:** Although the encoder adopts a VAE-style reparameterization and the model is repeatedly described as “variational”, the training objective contains only reconstruction terms (MSE and Cross-Entropy) and *no Kullback–Leibler divergence regulariser*. Without the KL term constraining the posterior toward a prior, the latent space is unregularised; strictly, the model behaves as a noise-injected deterministic auto-encoder rather than a genuine VAE, and the claimed benefits of “capturing uncertainty” are weakly substantiated.
- **Overstated Comparative Superiority:** The reported results support a more nuanced claim than uniform dominance. While the model leads on AUC-PR (0.89) and F1 (0.81), its ROC-AUC (0.85) is in fact the *lowest* among all graph baselines (0.91–0.93). The advantage is confined to precision–recall behaviour under imbalance; describing the model as comprehensively superior is not supported by its own ROC-AUC evidence, and the discrepancy is never reconciled.
- **Ambiguous Threshold Determination:** The decision threshold is specified inconsistently—via a statistical rule ($\mu + 2\sigma$ on validation loss) in the methodology, yet via an F1-maximising sweep in the experiments. The two procedures can yield materially different operating points, and if the threshold is in fact tuned on the test set, the reported F1 may be optimistically biased through label leakage. The paper does not clearly isolate a dedicated validation split for this calibration step.

B. Limitations Inherited from the GNN Family.

- **Computational Scalability:** The algorithm exhibits a time complexity bounded by $O(nE)$. In a global financial network encompassing billions of nodes and temporal edges, executing multi-head attention across heterogeneous neighbourhoods imposes severe GPU-memory bottlenecks, and full-graph training becomes infeasible

without sampling-based mini-batching such as ClusterGCN [Chiang et al. \(2019\)](#) or GraphSAINT [Zeng et al. \(2020\)](#).

- **Sub-optimal Layer Design Risks:** As highlighted by Ma et al. [Ma et al. \(2025\)](#), interleaving non-linear activations between message-passing layers can suppress the pairwise relational signal that simple, *linear*-propagation encoders preserve. If the HGAE relies on heavy intermediate non-linearities, it may be inadvertently degrading its own structural expressiveness—an effect compounded by its excessive reported depth.
- **The Static Graph Fallacy:** The formulation operates on a static graph snapshot G . Adversarial fraud, however, is dynamic: attackers continually mutate their activity patterns to evade detection. The absence of a temporal dimension renders the model blind to chronological behavioural shifts and velocity-based attacks (e.g. bursts of transactions in a short window).
- **Lack of Explanatory Power:** Financial regulation requires that automated decisions blocking a transaction be justifiable. The HGAE emits a binary flag derived from a continuous reconstruction loss but provides no mechanism to attribute that loss to a human-readable root cause, limiting its admissibility in compliance-sensitive settings.
- **Oversmoothing, Oversquashing, and Heterophily:** Deepening MPNN layers to capture long-range dependencies triggers *oversmoothing* (indistinguishable representations) and *oversquashing* (compression of exponentially growing receptive fields into fixed-size vectors) [Alon and Yahav \(2021\)](#). Moreover, standard GNNs assume homophily, whereas fraudsters deliberately transact with legitimate merchants to camouflage themselves, inducing *heterophily* [Luan et al. \(2024\)](#). The architecture offers no dedicated mechanism for such adversarially heterophilic connections.
- **Latent-Space Collapse:** A fundamental risk in purely unsupervised GAEs is representation collapse, in which the encoder maps disparate nodes to near-identical latent vectors, trivially minimising the loss without learning meaningful structure [Li et al. \(2024\)](#). While reparameterization injects noise, the model lacks contrastive regularisers (e.g. InfoNCE) that would actively separate dissimilar legitimate transactions, leaving it potentially fragile against sophisticated, camouflaged fraud [Li et al. \(2024\)](#).

4. Future Research Directions: Advancing Unsupervised Graph Learning

Guided by the limitations identified above, we outline five directions for advancing self-supervised and unsupervised graph learning in fraud detection. Each is anchored to a specific weakness of the HGAE.

- **Masked and Contrastive Graph Auto-Encoders (MGAE):** The frontier of unsupervised graph learning lies in masking and contrastive objectives [Tan et al. \(2023\)](#); [Li et al. \(2024\)](#). Addressing the *latent-collapse*

and *attribute-only reconstruction* limitations, future systems should adopt Masked Graph Auto-Encoders that randomly mask a large fraction (e.g. 15–70%) of edges or node features and force the model to reconstruct them, yielding far more resilient representations [Tan et al. \(2023\)](#). Coupling this with a contrastive objective (as in the Ir-GAE framework [Li et al. \(2024\)](#)) maximises agreement between augmented views of the same subgraph while actively pushing dissimilar normal transactions apart, directly mitigating representation collapse and strengthening robustness to camouflaged fraud.

- **Architectural Simplification for Scalability:** Paradoxically, as Ma et al. demonstrate [Ma et al. \(2025\)](#), future research should investigate *simplifying* the unsupervised encoder rather than deepening it. Removing non-linear activations between intermediate message-passing layers and applying orthogonal initialization lets a simple GAE implicitly capture higher-order common-neighbour heuristics while achieving state-of-the-art link prediction. This linear-propagation design is far cheaper to train, directly addressing the $O(nE)$ scalability bottleneck and the implausible-depth concern, and is well suited to billion-scale financial graphs when combined with sampling [Chiang et al. \(2019\)](#); [Zeng et al. \(2020\)](#).
- **Temporal Heterogeneous Graph Networks (T-HGNN):** To overcome the *static-graph fallacy*, future unsupervised models must move beyond single snapshots. Integrating recurrent or time-encoding components into the message-passing layers would allow the auto-encoder to reconstruct not only spatial topology and attributes but also chronological behavioural sequences, enabling detection of velocity-based and rapidly evolving fraud patterns [Majumder et al. \(2025\)](#).
- **Adaptive and Non-Parametric Thresholding:** The global $\mu + 2\sigma$ rule implicitly assumes a single, Gaussian error distribution shared by all entities, which is unrealistic: the “normal” reconstruction error of a high-spend customer differs from that of a low-spend one. Future work should explore *entity-conditioned* or *non-parametric* thresholds (e.g. per-segment quantiles or density-based scores), removing both the Gaussian assumption and the single-operating-point fragility highlighted in our critique.
- **Explainable Anomaly Attribution:** To satisfy regulatory demands for justifiable decisions, future models should pair the reconstruction-error detector with attribution mechanisms—for instance, decomposing the loss across node attributes and relations, or visualising the attention weights—so that a flagged transaction is accompanied by a human-readable rationale rather than an opaque score.

5. Conclusion

The Heterogeneous Graph Auto-Encoder represents a paradigm shift in financial fraud detection. By leveraging type-specific attention mechanisms and unsupervised reconstruction loss, it elegantly solves the class imbalance problem while fully utilizing the rich topological structures of financial

networks. Addressing its current limitations in scalability, temporal dynamics, heterophily assumptions, and interpretability will pave the way for next-generation, robust security systems.

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